



G.PULLA REDDY ENGINEERING COLLEGE (AUTONOMOUS): KURNOOL
DEPARTMENT OF EMERGING TECHNOLOGIES IN COMPUTER SCIENCE
B.TECH. – III SEMISTER (Scheme- 23)
LAB NAME(ADSA(P)) LABORATORY

```
else  
traversal(root);  
break;  
}  
printf("\n");  
menu();  
}  
}
```

EXPERIMENT-3

AIM:

Implement a program to sort the elements of an array using Heap sort technique.

Program :

```
// Heap Sort in C  
  
#include <stdio.h>  
  
#include <stdlib.h>  
  
// Function to swap the the position of two elements  
  
void swap(int *a, int *b) {  
  
    int temp = *a;  
  
    *a = *b;  
  
    *b = temp;  
  
}  
  
void heapify(int arr[], int n, int i) {  
  
    // Find largest among root, left child and right child  
  
    int largest = i;  
  
    int left = 2 * i + 1;  
  
    int right = 2 * i + 2;
```



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```
if (left < n && arr[left] > arr[largest])
```

```
largest = left;
```

```
if (right < n && arr[right] > arr[largest])
```

```
largest = right;
```

```
// Swap and continue heapifying if root is not largest
```

```
if (largest != i) {
```

```
swap(&arr[i], &arr[largest]);
```

```
heapify(arr, n, largest);
```

```
}
```

```
}
```

```
// Main function to do heap sort
```

```
// Main function
```

```
void heapSort(int arr[], int n)
```

```
{
```

```
int i;
```

```
// Build max heap
```

```
for ( i = n / 2 - 1; i >= 0; i--)
```

```
heapify(arr, n, i);
```



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```
for (i = n - 1; i >= 0; i--)  
{  
    swap(&arr[0], &arr[i]);  
  
    // Heapify root element to get highest element at root again  
    heapify(arr, i, 0);  
}  
}  
  
// Printing the array  
void printArray(int arr[], int n)  
{  
    int i;  
    for (i = 0; i < n; ++i)  
        printf("%d ", arr[i]);  
    printf("\n");  
}  
  
void main()  
{  
    //clrscr();  
  
    int *arr;  
  
    int j,n;  
  
    printf("Enter no. of elements:");
```



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```
scanf("%d", &n);

arr = (int*)malloc(n * sizeof(int));

if(arr == NULL)

printf("Memory not allocated.\n");

else

{

for (j = 0; j < n; ++j)

scanf("%d",&arr[j]);

}

heapSort(arr, n);

printf("Sorted array is: \n");

printArray(arr, n);

}
```

EXPERIMENT-4

AIM : Implement BFT and DFT for given graph, when graph is represented by a) Adjacency Matrix
b) Adjacency Lists

Program : BFT

a)Adjacency Matrix

```
#include<stdio.h>
#include <stdbool.h>
#define MAX 100
// BFS from given source s
void bfs(int adj[MAX][MAX], int V, int s) {

// Create a queue for BFS
```